

This phase of the lab setup transitions from the "Attacker" perspective to the "Defender" (Blue Team). We will set up the **Ubuntu Snort** VM to act as an Intrusion Detection System (IDS) that monitors the internal network for the very attack you just demonstrated.

1. Installing snort on Ubuntu

Even though your Kali machine has nmap, having it on your Ubuntu boxes is useful for internal auditing and connectivity testing.

1. Open the terminal on your **Ubuntu Snort** VM (\$10.0.0.13\$).
 2. Update the package list:
`sudo apt update`
 3. Install Snort:
`sudo apt install snort -y`
 4. Verify the installation:
`snort --version`
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2. Deploying Emerging Threats Rules

The emerging.rules.tar.gz file contains high-quality signatures used by security professionals to detect modern exploits like EternalBlue.

1. **Locate the archive:** Ensure emerging.rules.tar.gz is in your home directory.
 2. **Decompress the file:**
`tar -xvzf emerging.rules.tar.gz`
(This creates a folder, typically named rules.)
 3. **Copy to Snort Directory:**
`sudo cp rules/*.rules /etc/snort/rules/`
 4. **Copy Configuration files:** (If the archive included them)
`sudo cp rules/*.conf /etc/snort/`
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3. Configuring Snort (snort.conf)

You need to tell Snort which network to protect and which rule files to load.

1. **Open the configuration file:**
`sudo nano /etc/snort/snort.conf`
2. **Set the Network Variables:**
Find the section for ipvar (usually near the top) and modify them as follows.
Note: You requested 192.168.100.0/24 in your prompt, but since your lab uses 10.0.0.x, ensure your students understand

that HOME_NET must match the local subnet to trigger rules correctly.

Bash

```
ipvar HOME_NET 10.0.0.0/24 # Adjusted to match your lab's 10.0.0.x IP scheme
ipvar EXTERNAL_NET any
```

```
ipvar HTTP_SERVERS $HOME_NET
ipvar SMTP_SERVERS $HOME_NET
ipvar DNS_SERVERS $HOME_NET
ipvar FTP_SERVERS $HOME_NET
ipvar SQL_SERVERS $HOME_NET
```

3. Enable the Emerging Rules:

Scroll down to the "Step 7" area of the config file where rules are included. Add or uncomment the following lines:

Bash

```
include threshold.conf
include $RULE_PATH/emerging-activex.rules
include $RULE_PATH/emerging-attack_response.rules
include $RULE_PATH/emerging-botcc.rules
include $RULE_PATH/emerging-ciarmy.rules
include $RULE_PATH/emerging-compromised.rules
include $RULE_PATH/emerging-current_events.rules
#include $RULE_PATH/emerging-dns.rules
include $RULE_PATH/emerging-dos.rules
include $RULE_PATH/emerging-exploit.rules
include $RULE_PATH/emerging-malware.rules
include $RULE_PATH/emerging-netbios.rules
include $RULE_PATH/emerging-policy.rules
include $RULE_PATH/emerging-scan.rules
include $RULE_PATH/emerging-shellcode.rules
include $RULE_PATH/emerging-smtp.rules
#include $RULE_PATH/emerging-web_server.rules
```

4. Testing the Configuration

Before running the IDS, verify that there are no syntax errors in your snort.conf.

1. Run the Validation command:

```
sudo snort -T -c /etc/snort/snort.conf
```

2. Look for the Success Message:

If you see "Snort successfully validated the configuration!", you are ready.
